

Donghyung Lee

DevOps Engineer

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Introduction

Hello, I am Donghyung Lee, a DevOps Engineer.

Targeting 99.9% infrastructure availability, I have continuously improved operational efficiency by codifying every change as IaC and turning incident response into postmortem-driven assets. My greatest strength is going beyond routine operations to interrogate the meaning and purpose of each task and pursue a better direction.

I design and build AWS/GCP cloud and OpenStack on-premises PaaS infrastructure, operate Kubernetes (EKS, GKE, on-premises), apply Terraform and Ansible IaC, and build CI/CD pipelines on GitLab CI/CD, GitHub Actions, ArgoCD, and Jenkins. I've built and migrated monitoring/logging systems with PLG and ELK/EFK stacks, and handle operations automation directly in Python and Bash.

I've also been expanding my strengths into data engineering and AI/ML infrastructure — building data pipeline automation by combining BigQuery, Dataflow, Cloud Functions, and Cloud Scheduler (multi-source → BigQuery → Google Sheets, Terraform IaC) and deploying an internal LLM service (Ollama + Open WebUI) on GPU servers. Alongside that, I develop in-house automation / DevOps tools myself such as git-bridge, a Slack APK deploy bot, and static-file-server.

Across 3 companies, I have led AWS → GCP migration, hybrid architecture design, and on-premises PaaS deployment as the sole or primary owner, achieving 20% AWS cost reduction via hybrid transition, 50% reduction via GCP migration optimization, 50-67% deployment time cuts, and 90%+ automation of manual work.

I aim to continue contributing to infrastructure architecture advancement and organizational productivity improvement.

Strengths

Quick Understanding & Adaptation

- Quickly adapt to new environments and tools, contributing immediately to diverse projects

Systematic Documentation & Knowledge Sharing

- Document all work processes and share with team, contributing to organizational knowledge assets

Record-Based Continuous Improvement

- Manage all tasks including incident response, infra changes, and troubleshooting through records to prevent recurrence and drive improvement

Full-Stack Infrastructure Experience

- End-to-end infrastructure experience spanning cloud (AWS, GCP), on-premises, networking, CI/CD, and monitoring

Skills

Cloud	AWS, GCP, OpenStack
Container Orchestration	Kubernetes (EKS/GKE Multi-cluster Ops, On-premises HA with CNI/CSI Optimization)
Infrastructure as Code	Terraform, Ansible, Vagrant, Packer
CI/CD	GitHub Actions, GitLab CI/CD, ArgoCD, Jenkins
Monitoring & Logging	PLG Stack, ELK Stack, EFK Stack, ECK Operator, Thanos
Security	Vaultwarden, GitLab SSO (OIDC)
Automation	Go, Python, Bash Scripting
Database	MySQL, PostgreSQL, Redis, MongoDB
OS & Servers	Ubuntu, Rocky Linux, CentOS, Debian, GPU Server

Professional Experience

Phantom (Concrit Studio)

2024.10 ~ Present

DevOps Engineer

Dedicated DevOps infrastructure engineer designing, building, and operating AWS-based cloud and on-premises servers. Achieved both cost optimization and operational stability through hybrid cloud architecture design.

- AWS-on-premises hybrid cloud architecture design and construction
- GitOps CI/CD pipeline construction and enhancement based on GitLab CI / ArgoCD / Jenkins
- Built ELK Stack logging → migrated to EFK (70% resource reduction) → moved to ECK Operator + Elastic Stack 9.0 CRD-based declarative management (automatic TLS rotation, one-line CR spec.version rolling upgrades)
- Monitoring enhancement with kube-prometheus-stack (18 alert rules, 10 custom dashboards, 4 DB exporters)
- K8s cluster upgrade automation via Kubespray (9-step health checks, etcd backup/restore)
- Migrated 11 ingress-nginx instances to NGINX Gateway Fabric (NGF) (single control-plane + 11 Gateway CRs, wildcard TLS consolidation, zero-downtime cutover 11/11)
- Developed and released the open-source Helm chart nginx-gateway-cr
- Developed and released open-source Helm charts elasticsearch-eck / kibana-eck (based on ECK Operator, distributed via OCI + ArtifactHub, zero-downtime migration preserving CR/PVC names, HA rolling verified)
- Vaultwarden password management system (GitLab SSO, automated backup, team access control)
- GitLab-Google Drive document automation system
- Internal LLM service deployment (Ollama + Open WebUI)
- Slack APK distribution automation bot development and operation
- Developed and released Git repository mirroring tool (git-bridge) as open source
- GitLab, NAS, SVN account management automation via Bash Scripts
- Integrated application login environment using GitLab SSO
- Kubernetes storage system (NFS dynamic provisioning + Local Path Provisioner + in-house open-source static file server static-file-server)

Key Achievements

- 20% Monthly Infra Cost Reduction (\$60,000 → \$48,000, ~\$144K annual savings → reinvested in new projects)
- 67% Deployment Time Reduction (30min → 10min, 3x weekly deployments for faster market response)
- 90% Deployment Resource Efficiency (Enabling dev team to focus on core development)
- 90% Log Search Time Reduction
- 70% MTTR Improvement (99.9% Service Availability)

NerdyStar

2023.03 ~ 2024.07

DevOps Engineer

Led the large-scale AWS → GCP cloud migration at a game and blockchain startup, and rebuilt GitOps-based CI/CD pipelines.

- AWS → GCP large-scale cloud migration (Shared VPC + Workload Identity Federation for keyless GitHub Actions auth, zero-downtime DNS cutover via subdomain delegation, Cloud Armor + Cloud CDN, self-hosted NFS replacing Filestore)
- GitOps CI/CD pipeline construction based on GitLab CI / ArgoCD
- Certificate automation using Cert-Manager and Let's Encrypt
- Game/blockchain data collection automation pipeline built on Cloud Functions + Cloud Scheduler + Dataflow + BigQuery (MongoDB / CloudSQL / GA / Dune → BigQuery → Google Sheets, Terraform IaC)
- Integrated monitoring system based on PLG Stack (Prometheus, Loki, Grafana)
- GPU server infrastructure design and construction for AI/ML workloads

Key Achievements

- 50% Cloud Cost Reduction (\$10,000 → \$5,000/month)
 - 50% Deployment Time Reduction (40min → 20min)
 - 95% Rollback Time Reduction (30min → 1-2min)
 - 95% Data Collection Automation (0% data gap rate)
 - 99.9% Service Availability (99.5% → 99.9%)
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Iorchard

2022.04 ~ 2023.03

Infra Engineer

Built PaaS (Kubernetes + OpenStack + Ceph) infrastructure solutions for clients and provided technical support. Designed and built on-premises cloud infrastructure for financial and public sector clients, and conducted operations training.

- Customized PaaS solution design and construction (Kubernetes HA + OpenStack + Ceph)
- DP (Delivery Partner) Kubernetes operations training program design and delivery (6 months)
- Kubernetes certificate validity extension automation script (1yr → 10yr)
- OpenStack and Ceph storage cluster operations and optimization
- Regular cluster inspections and troubleshooting
- Ansible-based server provisioning automation eliminating configuration drift issues

Key Achievements

- 5 Cases Client PaaS Solution Deployments (99.9% Availability)
- 10x Certificate Renewal Cycle Extension (Annual → Once per 10 years)
- 50hrs Annual Ops Savings (Across 10 clients)

ETech System

2022.01 ~ 2022.04

System Engineer

- Hardware Server and SAN storage construction
- Transitioned to DevOps field for career growth

Opensource Projects

Kubernetes CRD

[k8s-namespace-sync](#) — Automatically sync Kubernetes namespace resources across clusters

[helios-lb](#) — Custom load balancer controller for Kubernetes

[network-policy-generator](#) — Kubernetes NetworkPolicy auto-generation controller

Helm Charts

[nginx-gateway-cr](#) — Tenant-level resource Helm chart for NGINX Gateway Fabric

[elasticsearch-eck](#) — Elasticsearch CR Helm chart for ECK Operator

[kibana-eck](#) — Kibana CR Helm chart for ECK Operator

[certmanager-letsencrypt](#) — Let's Encrypt resources Helm chart for cert-manager

GitHub Actions

[Compress Decompress Action](#) — Compress/decompress files in GitHub Actions workflows

[Go Git Commit Action](#) — Go-based Git commit creation and push

[Multi Git Mirror Action](#) — Automate mirroring across multiple Git repositories

[Image Tag Updater](#) — Automatically update image tags in YAML or Helm files

[Kube Diff Action](#) — Compare and report Kubernetes resource changes

[Helm Chart Release Action](#) — One-shot Helm chart release (package, gh-pages publish, OCI push)

[Go Kubebuilder Test Action](#) — kubebuilder operator test + manifests drift check

Ansible Galaxy

[Ansible K8s IAC Tool](#) — Automated installation collection for Kubernetes and IaC tools

[Ansible Kubectrl Krew](#) — kubectrl plugin Krew management role

[Ansible User Management](#) — Linux user and SSH key management role

DevOps Tools

[bash-pilot](#) — AI-powered Bash command assistant CLI tool

[git-bridge](#) — Git repository mirroring and sync CLI tool

[static-file-server](#) — Go-based static file server with directory UI, file preview, search, ZIP batch download

Chrome Extensions

[Dev Toolkit](#) — A collection of useful developer utilities

[QRify](#) — Quickly generate QR codes for URLs or text

Education

Chungnam National University - Public Administration (Transfer)	2018.03 ~ 2020.08
National Institute for Lifelong Education - Business Administration (Academic Credit Bank)	2016.09 ~ 2018.02

Certifications

Engineer Information Processing	2021.11
AWS Certified Solution Architect - Associate (Expired)	2021.11
Network Administrator Level 2	2021.10
Linux Master Level 2	2021.10
Computer Specialist Level 1	2021.04
TOEIC Speaking Test - 130/Intermediate Mid 3 (Expired)	2021.02

Study

T101(Terraform) Study	2023.08 ~ 2023.10
AEWS(EKS) Study	2023.04 ~ 2023.06
PKOS(Kubernetes) Study	2023.01 ~ 2023.02
Cloud Security Training	2022.02 ~ 2022.04
Security Solution Operation Specialist Training Program	2021.07 ~ 2021.12